

Two DFAs $M = (Q_M, \Sigma, \delta_M, s^M, F_M)$

$N = (Q_N, \Sigma, \delta_N, s^N, F_N)$

M is isomorphic to N if $\exists$ a bijection $F: Q_M \rightarrow Q_N$

s.t $F(s^M) = s^N$; $p \in F_M$ iff $F(p) \in F_N$

$F(\delta_M(p, a)) = \delta_N(F(p), a) \quad \forall p \in Q_M, \forall a \in \Sigma$

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Myhill - Nerode Theorem.

if M and N are any two DFAs with no inaccessible states s.t. $L(M) = L(N)$, then $M/\approx$ and $N/\approx$ are isomorphic.

Corollary.

The output of the collapsing algorithm is the **minimal DFA** for the set and it is **unique** up to isomorphism.

Let $M = (Q, \Sigma, \delta, s, F)$ be a DFA s.t. $L(M) = R$

M induces an equivalence relation $\equiv_M$ on Σ^* defined by

$$x \equiv_M y \text{ iff } \hat{\delta}(s, x) = \hat{\delta}(s, y)$$

Observation. $\equiv_M$ is an equivalence relation.

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Observation. $\equiv_M$ is an equivalence relation.

$\equiv_M$ also satisfies the following properties

(P1) It is a right Congruence: $\forall x, y \in \Sigma^*, \forall a \in \Sigma$

$$x \equiv_M y \Rightarrow xa \equiv_M ya.$$

Suppose $x \equiv_M y$. Then

$$\begin{aligned} \delta(s, xa) &= \delta(\hat{\delta}(s, x), a) \\ &= \delta(\hat{\delta}(s, y), a) \\ &= \hat{\delta}(s, ya) \end{aligned}$$

(P2) It refines R : $\forall x, y \in \Sigma^*$

$$x \equiv y \Rightarrow (x \in R \text{ iff } y \in R)$$

Since $\hat{\delta}(s, x) = \hat{\delta}(s, y)$

(P3) It is of finite index: there are only finitely many equivalence classes.

There is exactly one equivalence class corresponding to each $q \in Q$.

$$\{x \in \Sigma^* \mid \hat{\delta}(s, x) = q\}$$

Myhill - Nerode Relation

Let $R \subseteq \Sigma^*$ be a regular set.

An equivalence relation $\equiv$ on Σ^* is a Myhill - Nerode relation for R if it satisfies the following 3 properties

(P1) It is a right Congruence : $\forall x, y \in \Sigma^*, \forall a \in \Sigma$

$$x \equiv y \Rightarrow xa \equiv ya.$$

(P2) It refines R : $\forall x, y \in \Sigma^*$

$$x \equiv y \Rightarrow (x \in R \text{ iff } y \in R)$$

(P3) It is of finite index : There are only finitely many equivalence classes.

Observation.

Any DFA $M = (Q, \Sigma, S, \delta, F)$ with $L(M) = R$ defines a Myhill - Nerode relation $\equiv_M$:

Let $R \subseteq \Sigma^*$ and $\equiv$ be any Myhill-Nerode relation for R . From $\equiv$ we can construct a DFA $M_{\equiv}$ s.t.
 $L(M_{\equiv}) = R$.

For $x \in \Sigma^*$ let $[x] = \{y \mid y \equiv x\}$ be the equivalence classes of string x .

Note: By (P3) there are only finitely many equivalence classes.

$M_{\equiv} = (Q, \Sigma, S, \delta, F)$ where

$$Q = \{[x] \mid x \in \Sigma^*\}; \quad S = [\epsilon]; \quad F = \{[x] \mid x \in R\}$$

$$\delta([x], a) = [xa].$$

Question. if you choose another $y \in [x]$, can it be that $[xa] \neq [ya]$? No by right congruence.

Proposition 1. $x \in R$ iff $[x] \in F$.
 $\Rightarrow$ Definition of F .

$\Leftarrow$ Definition of F and (P2).

Lemma 1. $\hat{S}([x], y) = [xy]$.

Proof. Induction on $|y|$

Base Case: $\hat{S}([x], \epsilon) = [x] = [x\epsilon]$

Induction Step:

$$\begin{aligned}\hat{S}([x], ya) &= S(\hat{S}([x], y), a) \\ &= S([xy], a) \quad (\text{Induction hypothesis}) \\ &= [xya] \quad (\text{Definition of } S)\end{aligned}$$

Theorem. $L(M_{\equiv}) = R$

Proof

$$x \in L(M_{\equiv}) \Leftrightarrow \hat{S}([\epsilon], x) \in F$$

$$\Leftrightarrow [x] \in F \quad (\text{Lemma 1})$$

$$\Leftrightarrow x \in R \quad (\text{Proposition 1})$$

Lemma 2. Let $\equiv$ be a Myhill-Nerode relation for $R \subseteq \Sigma^*$. if we apply the construction $\equiv \mapsto M_{\equiv}$ and then apply the construction $M \mapsto \equiv_M$ to the result, the resulting relation $\equiv_{M_{\equiv}}$ is identical to $\equiv$

Proof.

Let $M_{\equiv} = (Q, \Sigma, \delta, \delta_0, F)$

For any $x, y \in \Sigma^*$

$$\begin{aligned}
 x \equiv_{M_{\equiv}} y & \text{ iff } \hat{\delta}(\delta_0, x) = \hat{\delta}(\delta_0, y) \text{ (Defn of } \equiv_{M_{\equiv}}) \\
 & \text{ iff } \hat{\delta}([\epsilon], x) = \hat{\delta}([\epsilon], y) \text{ (Defn of } \delta) \\
 & \text{ iff } [x] = [y] \text{ (By Lemma 1)} \\
 & \text{ iff } x \equiv y
 \end{aligned}$$

Lemma 3. Let M be a DFA for R (with no inaccessible states).

If we apply the construction $M \mapsto \Xi_M$ and then apply

$\Xi \mapsto M_\Xi$, the resulting DFA M_{Ξ_M} is isomorphic to M

Proof. Let $M = (Q, \Sigma, \delta, s, F)$ and

$$M_{\Xi_M} = (Q', \Sigma, \delta', s', F')$$

$$[x] = \{y \mid y \Xi_M x\} = \{y \mid \hat{\delta}(s, y) = \hat{\delta}(s, x)\}$$

$$Q' = \{[x] \mid x \in \Sigma^*\} \quad s' = [\epsilon] \quad F' = \{[x] \mid x \in R\}$$

$$\delta'([x], a) = [xa]$$

Consider the function $f: Q' \rightarrow Q$ defined as

$$f([x]) = \hat{\delta}(s, x)$$

$[x] = [y]$ iff $\hat{\delta}(s, x) = \hat{\delta}(s, y)$ (From defn of Ξ_M)
So f is well-defined and one-to-one.

f is onto since M has no inaccessible states.

claim. M_{Ξ_M} and M are isomorphic under f .

Consider the function $F: Q' \rightarrow Q$ defined as
$$F([x]) = \hat{S}(s, x)$$

claim. $M \equiv_M Q$ and M are isomorphic under F .
To show.

$$(1) F(s') = s$$

$$(2) F(S'([x], a)) = S(F([x]), a)$$

$$(3) [x] \in F' \text{ iff } F([x]) \in F$$

$$\begin{aligned} (1). \quad F(s') &= F([e]) && (\text{Defn of } s') \\ &= \hat{S}(s, e) && (\text{defn of } F) \\ &= s && (\text{Defn of } \hat{S}) \end{aligned}$$

$$\begin{aligned} (2). \quad F(S'([x], a)) &= F([xa]) && (\text{Defn of } S') \\ &= \hat{S}(s, xa) && (\text{Defn of } F) \\ &= S(\hat{S}(s, x), a) && (\text{Defn of } \hat{S}) \\ &= S(F([x]), a) && (\text{Defn of } F) \end{aligned}$$

$$\begin{aligned} (3). \quad [x] \in F' &\text{ iff } x \in R && (\text{Defn of } F \text{ and prop 2}) \\ &\text{ iff } \hat{S}(s, x) \in F && (\text{since } L(M) = R) \\ &\text{ iff } F([x]) \in F && (\text{Defn of } F). \end{aligned}$$

Myhill-Nerode Relation

Let $R \subseteq \Sigma^*$ be a regular set.

An equivalence relation $\equiv$ on Σ^* is a Myhill-Nerode relation for R if it satisfies the following 3 properties

(P1) It is a right Congruence : $\forall x, y \in \Sigma^*, \forall a \in \Sigma$

$$x \equiv y \Rightarrow xa \equiv ya.$$

(P2) It refines R : $\forall x, y \in \Sigma^*$

$$x \equiv y \Rightarrow (x \in R \text{ iff } y \in R)$$

(P3) It is of finite index : There are only finitely many equivalence classes.

Observation.

Any DFA $M = (Q, \Sigma, S, \delta, F)$ with $L(M) = R$ defines a Myhill-Nerode relation $\equiv_M$:

A relation $\equiv_1$ refines another relation $\equiv_2$ if

$$\equiv_1 \subseteq \equiv_2.$$

That is, $\forall x, y$ if $x \equiv_1 y$ then $x \equiv_2 y$.

For equivalence relation: $\forall x \quad [x]_1 \subseteq [x]_2$

"Refinement" - relation between equivalence relations is

- reflexive

- transitive (if $\equiv_1$ refines $\equiv_2$ and $\equiv_2$ refines $\equiv_3$ then $\equiv_1$ refines $\equiv_3$).

- antisymmetric (if $\equiv_1$ refines $\equiv_2$ and $\equiv_2$ refines $\equiv_1$ then $\equiv_1$ & $\equiv_2$ are the same)

if $\equiv_1$ refines $\equiv_2$ then $\equiv_1$ is the finer and

$\equiv_2$ is the coarser of the two relations.

There is always a "finest" and "coarsest" equivalence rel. on any set U .

Let $R \subseteq \Sigma^*$ (not necessarily regular)

Define an equivalence relation $\equiv_R$ on Σ^* as follows:

$$x \equiv_R y \text{ iff } \forall z \in \Sigma^* (xz \in R \text{ iff } yz \in R)$$

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Lemma. Let $R \subseteq \Sigma^*$ (R need not be regular)

The relation $\equiv_R$ is a right congruence refining R and it is the coarsest such relation on Σ^* .

Proof.

(1) $\equiv_R$ is a right congruence

In the definition of $\equiv_R$, take $z = aw$, $a \in \Sigma$, $w \in \Sigma^*$

$$\begin{aligned} x \equiv_R y &\Rightarrow \forall a \in \Sigma \forall w \in \Sigma^* (xaw \in R \text{ iff } yaw \in R) \\ &\Rightarrow \forall a \in \Sigma (xa \equiv_R ya) \end{aligned}$$

(2). $\equiv_R$ refines R .

In the definition of $\equiv_R$ take $z = \epsilon$.

$$x \equiv_R y \Rightarrow (x \in R \text{ iff } y \in R)$$

Coarsest Relation.

Let $\equiv$ be any other relation (Satisfies P_1, P_2)

$$x \equiv y$$

$$\Rightarrow \forall z (xz \equiv yz) \quad (\text{By induction on } |z| \text{ using } P_1)$$

$$\Rightarrow \forall z (xz \in R \text{ iff } yz \in R) \quad (\text{Property } P_2)$$

$$\Rightarrow x \equiv_R y \quad (\text{Definition of } \equiv_R)$$

Myhill-Nerode Theorem. Let $R \subseteq \Sigma^*$. The following statements are equivalent.

1. R is regular

2. There exists a Myhill-Nerode relation for R

3. The relation $\equiv_R$ is of finite index.

Proof.

$1 \Rightarrow 2$. Given a DFA M s.t. $L(M) = R$, the construction

$M \mapsto \equiv_M$ produces a Myhill-Nerode relation for R .

$2 \Rightarrow 3$. Any Myhill-Nerode relation $\equiv$ for R is of finite index.

By Lemma, $\equiv$ refines $\equiv_R$. So $\equiv_R$ has finite index.

$3 \Rightarrow 1$. If $\equiv_R$ is of finite index then it is a Myhill-Nerode relation for R .

The construction $\equiv \mapsto M_\equiv$ generates a DFA s.t. $L(M_\equiv) = R$.

Suppose $M = (Q, \Sigma, \delta, s, F)$ is a DFA where $L(M) = R$
and $M = M/\approx$ (Collapsed).

That is,

$$p \approx q \text{ iff } \forall x \in \Sigma^* (\hat{\delta}(p, x) \in F \text{ iff } \hat{\delta}(q, x) \in F)$$

is identity on Q and M has no inaccessible states.

Claim. $\equiv_M$ the Myhill-Nerode relation corresponding to M is same as $\equiv_R$.

Proof.

$$x \equiv_R y \Leftrightarrow \forall z \in \Sigma^* (xz \in R \text{ iff } yz \in R) \quad (\text{Defn of } \equiv_R.)$$

$$\Leftrightarrow \forall z \in \Sigma^* (\hat{\delta}(s, xz) \in F \text{ iff } \hat{\delta}(s, yz) \in F) \quad (\text{Defn of acceptance})$$

$$\Leftrightarrow \forall z \in \Sigma^* (\hat{\delta}(\hat{\delta}(s, x), z) \in F \text{ iff } \hat{\delta}(\hat{\delta}(s, y), z) \in F) \quad (\text{Claim-Exercise})$$

$$\Leftrightarrow \hat{\delta}(s, x) \approx \hat{\delta}(s, y) \quad (\text{Definition of } \approx)$$

$$\Leftrightarrow \hat{\delta}(s, x) = \hat{\delta}(s, y) \quad (\text{Since } M \text{ is collapsed})$$

$$\Leftrightarrow x \equiv_M y \quad (\text{Definition of } \equiv_M)$$